

SUPPLEMENTARY MATERIAL

A COMPLETION: ALGORITHM AND DERIVATION

One outer step. Both views are linearized at the current state. Their mean block curvature defines the whitening metric. The solver constructs a measured basis Q , solves its mean quadratic, and asks the network for additional directions. The projected directions define a small two-view problem; its solution is refined and mapped back to physical coordinates before taking a finite step:

1. Form $g = (g_a + g_b)/2$ and $H = (H_a + H_b)/2$. Build Q from gradient, history, graph-weighted and Krylov directions, caching $Y_\nu = H_\nu Q$.
2. Compute $A = Q^\top H Q$, $c = A^{-1} Q^\top g$, $d_0 = -Qc$ and $e = g - H Q c$.
3. Generate V with the shared network and set $Z = \text{orth}((I - Q A^{-1} Q^\top H) V)$, removing dependent columns.
4. Evaluate $H_a Z$, $H_b Z$ and solve Eq. (3). If its numerical checks fail, retain d_0 .
5. Apply the configured mean-CG refinement. Map the resulting direction to physical coordinates and form the bounded acceleration in Eq. (4).
6. Compare straight and curved trials at each backtracking time, then accept using the mean observation objective. Store the accepted displacement as history.

Preserving the base solve. Since $A = Q^\top H Q$,

$$Q^\top H (I - Q A^{-1} Q^\top H) = 0. \quad (6)$$

Orthogonalizing the projected columns preserves this nullspace. Hence $Q^\top H Z = 0$, and $Q^\top (g + H(d_0 - Z\alpha)) = 0$ for every α . The implementation additionally removes roundoff in the span of HQ . Each learned proposal is thus used to extend a stationary base solution.

Scalar dual. Let $C_w = wC_a + (1-w)C_b$ and $b_w = wb_a + (1-w)b_b$. Introducing an epigraph for the two quadratic changes gives nonnegative multipliers summing to one. Eliminating α yields

$$\max_{0 \leq w \leq 1} \phi(w), \quad \phi(w) = -\frac{1}{2} b_w^\top C_w^{-1} b_w, \quad \alpha_w = C_w^{-1} b_w. \quad (7)$$

The positive-definite reduced matrices make the primal convex, and strict epigraph feasibility gives strong duality. With $D = C_a - C_b$ and $b_\Delta = b_a - b_b$,

$$\phi'(w) = -b_\Delta^\top \alpha_w + \frac{1}{2} \alpha_w^\top D \alpha_w. \quad (8)$$

The solver checks the endpoints and bisection an interior root. Choosing $\alpha = 0$ gives zero change in both views, so the exact solution satisfies $\Delta_a(\alpha^*), \Delta_b(\alpha^*) \leq 0$. This property applies to completion before the subsequent mean-CG refinement.

Common curvature. If $C_a = C_b = C$, write $m = (b_a + b_b)/2$ and $\delta = (b_a - b_b)/2$. The objective becomes

$$\frac{1}{2} \alpha^\top C \alpha - m^\top \alpha + |\delta^\top \alpha|. \quad (9)$$

Its optimality condition is $C\alpha - m + \eta\delta = 0$, with $\eta \in [-1, 1]$ in the subgradient of the absolute value. Substitution gives Eq. (5). At an interior dual optimum, training differentiates the stationarity equation implicitly; endpoint solutions follow their active branch.

B NETWORK, TRAINING AND SOLVER CONFIGURATION

Features. The 20 carriers contain e , $g_a - g_b$, $(g_a + g_b)/2$, the previous accepted displacement, and eight cached curvature responses from each view. Missing responses and initial history are zero. Each carrier is normalized by its full-state Euclidean norm. For every block, the network receives 400 entries of the carrier Gram matrix, four block descriptors and 32 probe summaries.

The block descriptors are its fraction of the state dimension, residual-carrier energy, normalized graph degree and an indicator for nonzero degree. For probe q_j , define

$$s_j = \sqrt{\frac{1}{2}(\|Y_a[:, j]\|^2 + \|Y_b[:, j]\|^2)}, \quad y_{\nu j} = Y_\nu[:, j]/s_j. \quad (10)$$

The four summaries per probe are the blockwise sums of $q_j \odot (y_{aj} + y_{bj})/2$, y_{aj}^2 , y_{bj}^2 and $y_{aj} \odot y_{bj}$. Both views use the same s_j , preserving their relative response magnitude. Norm denominators have a numerical floor.

Network. Two width-48 linear-LayerNorm-SiLU layers embed the 436 features. Two residual message rounds concatenate each block’s hidden state with the mean of its graph neighbors, then apply a width-48 linear-LayerNorm-SiLU map. The head emits $8(1 + 2 \cdot 20) = 328$ values per block: residual coefficients, local carrier coefficients and global coefficients averaged over blocks. All blocks use the same 49,096 parameters. Eq. (2) gives the output construction.

Training. Local subspace pretraining uses synthetic matrix, sensor, SE3 and bundle-adjustment problems, plus radiometric calibration on three HPatches sequences: `i_autannes`, `i_bologna` and `i_books`. It evaluates initialization seeds 17, 29 and 43 using separate validation problems. The selected local checkpoint (seed 17, update 120) initializes both trajectory runs. Trajectory training uses 30 synthetic SE3 and 30 bundle-adjustment instances, with six separate validation instances from each family. Its objective is

$$\mathcal{L}_T = \frac{\bar{F}(x_T) - \bar{F}(x_{LM,5})}{\max\{\bar{F}(x_0) - \bar{F}(x_{LM,5}), 10^{-3}\bar{F}(x_0), 10^{-30}\}}. \quad (11)$$

Training uses AdamW with learning rate 10^{-4} , weight decay 10^{-5} and gradient clipping at norm 2. The trajectory protocol fixes two training-order seeds, 17 and 29, each starting from the same pretrained weights. Each run receives 60 updates: the first 40 unroll two steps and the last 20 unroll five. Both runs are evaluated at updates 0, 20, 40 and 60 on the same 12 validation problems, always using five-step inference. Selection minimizes the mean normalized observation loss across these eight evaluations. Seed 29 at update 40 attains the minimum, 0.003151903, versus 0.003220161 at the common initialization. This checkpoint is still in the two-step training phase; all reported inference uses five steps. The seed pair defines the trajectory-stage selection budget, rather than a test-time average over models. Every paper result uses this one checkpoint. Recovery truth does not enter its training or selection; the supplementary package records all eight validation scores.

Numerics and refinement. Subspace arithmetic uses float64. The measured and completion budgets are eight directions each, with rank tolerance 10^{-9} and completion residual tolerance 10^{-7} . A twice-reorthogonalized basis removes dependent directions. Measurement graphs, Broyden and TUM use two mean-CG corrections; the remaining main configurations use none. Curvature is Gauss-Newton for residual objectives and a positive-definite shifted Hessian for scalar objectives. The block metric is formed from mean curvature, with eigenvalue floor $32\epsilon_{\text{mach}} \max_i |N_{ii}|$.

Finite step. The acceleration uses

$$r_{vv} \approx \frac{2[r(x + hv) - r(x) - hJv]}{h^2}, \quad h = 0.02, \quad (J^\top J + \lambda D)a = -J^\top r_{vv}. \quad (12)$$

Here D is the floored diagonal of $J^\top J$. Backtracking tests $t = 1, 1/2, \dots, 1/32$ with Armijo coefficient 10^{-4} . Acceptance requires positive predicted reduction and an actual-to-predicted ratio above 10^{-3} . Relative damping starts at 10^{-3} , decreases by a factor 0.3 after a full step with ratio above 0.75, increases by 3 below 0.25, and by 10 on rejection, within $[10^{-9}, 10^6]$. The curve envelope uses tolerance 10^{-14} times the mean absolute straight-trial cost, floored at 10^{-30} .

C WHITENING EQUIVARIANCE AND EQUIVALENT EXECUTION

Proof of Proposition 1. Let W and W' whiten the same positive-definite block metric N , with $W' = WO$ and block-diagonal orthogonal O . A physical gradient γ_ν , curvature B_ν and displacement history h are represented as

$$g_\nu = W^\top \gamma_\nu, \quad H_\nu = W^\top B_\nu W, \quad h_W = W^{-1} h. \quad (13)$$

Replacing W gives $g'_\nu = O^\top g_\nu$, $H'_\nu = O^\top H_\nu O$ and $h_{W'} = O^\top h_W$. Block energies and graph-weighted scalar seeds are unchanged. Ordered orthogonalization and Krylov actions therefore give $Q' = O^\top Q$ and $Y'_\nu = O^\top Y_\nu$. Carrier norms, block Gram entries and probe summaries are invariant, so the hidden states and predicted scalar coefficients are identical. Carrier mixing gives $V' = O^\top V$.

The curvature-orthogonal projector transforms by conjugation. Thus $Z' = O^\top Z$, the reduced matrices and right-hand sides are unchanged, and the same α gives $d' = O^\top d$. Mean-CG refinement obeys the same transformation. Mapping back yields $W'd' = Wd$. Physical acceleration, trial costs and damping consequently receive the same inputs; induction gives the same trajectory. The argument fixes the physical metric, block partition and graph, and assumes unchanged rank decisions. Floating-point branches are checked separately in the experiment below.

Cholesky execution. Symmetric whitening takes $W = N^{-1/2}$. For $N = LL^\top$, the equivalent factor is $W' = L^{-\top}$. Gradients use $L^{-1}\gamma$, physical displacements use $L^{-\top}d$, and history uses $L^\top h$. This distinction between gradient and displacement maps is essential. The implementation enables Cholesky for a single full block when $N - \tau I$ is positive definite, using the original floor τ . Other layouts and clipped metrics keep the symmetric implementation.

For a full mean block $N = (B_a + B_b)/2$, the whitened views satisfy $H_a + H_b = 2I$. After forming one view, the second is obtained by subtraction. Paired eigen and paired Cholesky controls separate this reuse from the factorization change.

Table 3. Full five-step time in seconds, medians of three rotated serial repeats after one warmup. All routes share the original network and recorded host/backend. The last two cases confirm a frozen implementation. Ratios divide original time by Cholesky time. Non-stereo cases take the original fallback. The full-block stereo cases have zero exact minimax correction; these timings isolate equivalent execution and whitening cost.

Case	Original	Paired eigen	Cholesky	Ratio
Biased graph	0.0251	0.0247	0.0253	0.99×
Multiframe	3.0780	3.0190	3.2319	0.95×
Stereo A	7.9020	6.6296	2.7307	2.89×
Low light	2.4179	2.4865	2.4280	1.00×
SIDD	0.8425	0.8729	0.8691	0.97×
Stereo B	9.3961	7.5215	2.9686	3.17×
Stereo C	9.0197	7.9213	2.9613	3.05×

Verification and timing. The pilot uses five cases; two further stereo cases test the frozen implementation. Each route receives one warmup and three rotated serial repeats on an Intel Core Ultra 7 155H, Windows 11, Python 3.13.7 and PyTorch 2.7.0.dev20250310, with one numerical thread. Timings cover the complete five-step solve with prepared problems and models. The 84 endpoints are rescored with zero discrepancy; the largest relative endpoint difference is 1.35×10^{-11} . All 420 step records preserve acceptance, step scale, damping, measured rank and curve selection.

The three stereo cases show 2.89–3.17× acceleration. In this full-block setting the mean whitened curvature is I and the base already contains its optimum $d_0 = -g$. Any nonzero increment raises the mean quadratic by $\|u\|^2/2$, so exact minimax adds no further correction. This experiment isolates execution equivalence and whitening cost; it is not intended to measure acceleration of the learned-completion component. Learned-direction gains are measured separately against random completion in Section 4.4.

Additional equivalence cases. We compare one five-step solve per implementation on 30 stereo crops from 15 Middlebury scenes, using two fixed crops per scene and the same weights. Relative endpoint differences stay below 2.19×10^{-10} and recovery errors differ by less than 9.86×10^{-12} pixels. All 150 paired steps preserve the five control decisions above. These runs check numerical equivalence only.

D TASKS AND COMPARISON PROTOCOL

Shared problems. Within each comparison, methods receive the same observations, masks, variable parameterization and initial state. The primary quality budget is five outer steps. Internal rejection, damping retries and line searches follow each method’s implementation. The separate Jacobian-budget study is specified in Appendix H. Every SETSUNET run uses the same checkpoint.

Task metrics and inputs. Measurement graphs use state RMSE, Broyden uses residual RMS, and SE3/TUM use aligned translation RMSE in meters. HPatches pair and multiframe alignment use corner/reprojection error in pixels. Stereo uses disparity error in pixels. Restoration uses PSNR in dB. Results retain their task units; PSNR differences are additive, while reported relative error reductions specify whether they average per-instance ratios or compare group means.

The expanded visual protocols use HPatches, GoPro, LOL-v2, SIDD and Middlebury (Balntas et al., 2017; Nah et al., 2017; Yang et al., 2021; Abdelhamed et al., 2018; Scharstein et al., 2014):

- HPatches uses all 116 sequences and 580 pairs with a fixed feature front end; multiframe problems share this preparation.
- Known-blur recovery uses 64×64 regions from 66 frames in 11 GoPro videos, with a controlled blur operator. Low light uses 32×32 regions from 100 LOL-v2 test images.
- SIDD uses all 40 validation source images, four predefined standard blocks per image and one fixed 64×64 region per block. These three restoration groups use 16×16 tiled inference.
- Stereo uses 15 Middlebury scenes and 12 predefined crop positions per scene. These image experiments compare recovery optimizers on the stated regions, rather than full-resolution dataset leaderboards.

The expansion also includes 64 instances in each of four synthetic groups and 60 TUM windows from three sequences (Sturm et al., 2012). Shared preparation fails for one HPatches pair, its multiframe sequence and 13 TUM windows, leaving 1,503 cases; no replacements are added.

Baseline implementations. HSLM follows the authors’ formulation with the small-dimension extension used in these experiments. Ceres LM executes the native library trust-region solver with DENSE_QR, a five-iteration limit and one numerical thread. Its function tolerance is 10^{-14} and gradient/parameter tolerances are 10^{-12} . Dogbox calls SciPy’s rectangular trust-region solver with the exact shared residual/Jacobian interface, Jacobian-based scaling and 10^{-13} stopping tolerances; an iteration callback stops after five updates. Sparse Jacobians retain their sparse representation. Both libraries retain natural early termination.

VeLO uses the released HyperV2 numerical code and unchanged official meta-weights. The primary transfer flattens the physical state into one tensor, declares horizon five and executes five updates. It receives the shared scalar objective and gradient without task-fitted normalization or learning rates. The same configuration is used across all fixed-step comparisons. This evaluates transfer to inverse problems.

Reporting. The five comparison methods are SETSUNET, HSLM, Ceres LM, Dogbox and VeLO. Each scored case has all five endpoints. For error metrics $a, b \geq 0$, a practical tie satisfies $|a - b| \leq \max\{0.02 \min(a, b), 10^{-10} + 10^{-6} \max(a, b)\}$. The numerical floor prevents roundoff near zero from producing artificial wins. For PSNR scores p_a, p_b , a tie satisfies $|p_a - p_b| \leq 20 \log_{10}(1.02)$, corresponding to a 2% RMSE ratio at a common intensity range. Table 1 applies the same margin to each group mean and its best value; for mean PSNR this is a geometric-mean RMSE ratio. Larger advantages count as wins. This is a descriptive convention, not a test of statistical or perceptual significance. The two 32-case cohorts retain their distinct construction and selection histories.

E PER-TASK RESULTS AND COMPARISON COVERAGE

Table 4 reports the 18-task heterogeneous benchmark. The same shared network is used for each formulation and state dimension. Table 5 identifies the additional studies; all five methods are evaluated on every retained case.

Table 4. Five-step recovery on the 18-task heterogeneous benchmark. Parentheses give state dimensions. RMSE and pixel errors are lower-is-better; PSNR is higher-is-better.

Task	Problem (dimension)	Metric	SETSUNET	HSLM	Ceres LM	Dogbox	VeLO
T01	Biased graph (48)	RMSE	0.1165	0.6537	0.2068	0.2068	0.7400
T02	Biased graph (96)	RMSE	0.2321	0.7267	0.2857	0.2857	0.7643
T03	SE3 (18)	m	0.0014	0.0026	0.0014	0.0014	0.0039
T04	Multiframe (16)	px	2.5350	3.6085	3.5863	3.7153	3.5862
T05	Multiframe (16)	px	0.1637	0.2540	0.2116	0.1849	0.2113
T06	Known blur (256)	dB	40.1678	34.6637	40.1625	40.1625	33.0457
T07	Low light (786)	dB	8.2994	8.2896	8.2192	8.2480	8.2079
T08	Low light (786)	dB	8.3023	8.2751	8.2210	8.2841	8.2091
T09	SIDD (770)	dB	34.1144	33.2233	34.1158	34.1152	34.2352
T10	Stereo (1536)	px	0.1884	0.3941	0.4047	0.3501	0.4047
T11	Biased graph (48)	RMSE	0.1572	0.7118	0.1978	0.1978	0.7740
T12	Biased graph (96)	RMSE	0.2363	0.6827	0.2910	0.2910	0.7244
T13	SE3 (30)	m	0.0038	0.0046	0.0038	0.0038	0.0040
T14	Multiframe (40)	px	11.7885	14.7951	15.2581	16.6917	28.6592
T15	Known blur (256)	dB	53.4392	53.0480	53.4322	53.4322	47.2617
T16	Low light (786)	dB	8.0832	8.0402	8.0253	8.0547	8.0252
T17	SIDD (770)	dB	35.4657	35.0917	35.4655	35.4651	35.4356
T18	Stereo (1536)	px	0.2503	0.2933	0.3222	0.3020	0.3222

Table 5. Comparison coverage. Common preparation failures affect every method. The complementary suite combines 32 extension cases and 32 biased graphs, retaining cohort identities. The derivative study counts VeLO gradients separately from Jacobians.

Study	Scored/reg.	Budget	Methods
Heterogeneous tasks	18/18	5 steps	All five
Complementary cases	64/64	5 steps	All five
Large collections	1,503/1,518	5 steps	All five
Graph and multiframe	20/20	5 steps	All five
Five-target multiframe	7/8	5 steps	All five
Optimizer transfer	11/11	5 steps	All five
Derivative budget	60/60	21 derivatives	All five

The 64-case complementary suite combines twelve additional synthetic cases, twenty public-image problems and 32 further biased graphs. Figure 2 uses the full common 64-case denominator for each comparison and the 2% practical-tie rule.

F EXPANDED BENCHMARKS AND TRANSFER

Main-text Table 1 reports all 11 groups and all five methods on the same 1,503 prepared cases. Table 6 gives paired differences against the four comparators. The 10,000 bootstrap draws resample source sequences, images, scenes or synthetic seeds, preserving correlated crops within each source. Intervals are descriptive and have no simultaneous multiple-comparison correction.

Table 6. Mean recovery improvement and descriptive 95% source-cluster intervals. Positive favors SETSUNET. Units follow Table 1; restoration differences are in dB.

Group	HSLM	Ceres LM	Dogbox	VeLO
HPatches pairs	+10.3 [-0.111, +30.9]	-0.0381 [-0.175, +0.0456]	-0.0361 [-0.169, +0.0434]	+24.6 [-0.0399, +73.7]
Multiframe	+2.74 [-0.276, +7.17]	+3.49 [+0.17, +8.2]	+1.23 [-1.14, +4.41]	+12.3 [+3.72, +23]
Balanced graph	+0.447 [+0.399, +0.494]	-0.0989 [-0.116, -0.0819]	-0.0989 [-0.116, -0.0822]	+0.621 [+0.603, +0.637]
Biased graph	+0.343 [+0.291, +0.391]	+0.113 [+0.103, +0.124]	+0.113 [+0.103, +0.124]	+0.495 [+0.476, +0.515]
Broyden	+0.182 [+0.166, +0.201]	-5.37e-06 [-5.81e-06, -4.92e-06]	-5.37e-06 [-5.81e-06, -4.91e-06]	+0.522 [+0.492, +0.554]
SE3	+0.00406 [+0.00334, +0.00482]	-0.0015 [-0.00208, -0.000968]	-0.0015 [-0.00209, -0.000973]	+0.00446 [+0.00336, +0.00566]
TUM	+0.00696 [+0.00119, +0.0165]	-0.0225 [-0.0621, -0.00318]	-0.0226 [-0.0625, -0.00318]	+0.00859 [+0.00428, +0.0143]
Known blur	+1.77 [+1.67, +1.88]	-0.00067 [-0.00189, +0.000726]	-0.00067 [-0.00191, +0.00073]	+4.27 [+3.83, +4.75]
Low light	+0.113 [+0.0938, +0.133]	+0.118 [+0.0942, +0.141]	+0.0665 [+0.0549, +0.0787]	+0.0583 [+0.0472, +0.0693]
SIDD	+0.0876 [+0.0187, +0.206]	+0.0179 [+0.0139, +0.0222]	+0.00763 [+0.00577, +0.0097]	-0.0496 [-0.0836, +0.00719]
Stereo	+0.273 [+0.178, +0.388]	+0.347 [+0.233, +0.48]	+0.278 [+0.18, +0.396]	+0.347 [+0.234, +0.479]

The multiframe and stereo group means favor SETSUNET over all four comparators. The multiframe intervals against HSLM and Dogbox include zero, so these mean differences alone do not establish stable advantages across source sequences. The balanced-graph and pose results favor the classical library solvers. SIDD means are practically tied under the declared margin. Table 1 reports each group separately; the intervals retain their original metric differences.

Additional comparisons. The 20-case study contains fourteen synthetic problems and six public multiframe scenes. SETSUNET has win/tie/loss counts of 18/0/2, 12/2/6, 12/1/7, 16/1/3 against HSLM, Ceres LM, Dogbox and VeLO, respectively. The eleven-case transfer study gives 8/2/1, 3/6/2, 3/6/2, 8/2/1 in the same order. Both use the 2% margin and the same primary VeLO configuration.

Seven of the eight five-target multiframe scenes pass shared preparation. Mean errors are 8.5793, 8.3740, 8.3206, 7.9846, 180.5865 pixels for SETSUNET, HSLM, Ceres LM, Dogbox and VeLO, respectively. This larger observation graph preserves the same method definitions and five-step budget; all five methods use the same seven scenes.

G DIRECTION COMPARISONS, COORDINATION AND SIZE TRANSFER

Paired graph construction. The paired direction study uses 16 new seeds, 91914001–91914016, with 96 variables. Edges connect $(i, i + 1)$ and $(i, i + 3)$ where valid, with anchors $0.1I$. Truth is a sinusoid with Gaussian perturbations of standard deviation 0.2. Two Gaussian observation draws are generated once per scene and reused across strengths. View 0 has noise standard deviation 0.01. The noisy-view sweep gives view 1 standard deviation $0.01 + 0.09s$; the bias sweep keeps both at 0.01 and adds $0.12s \sin(t)$ to view 1, where measurement rows span $t \in [0, 4\pi]$. Strengths are $s = 1/3, 2/3, 1$, plus one common balanced condition, producing 112 conditions.

Direction and coordination controls. The reported direction-source comparison uses learned completion and three equal-rank random completions with the same minimax coefficients and CG2 refinement. The coordination comparison uses the same learned directions with mean or minimax coefficients, separately with CG2 and without CG. All reported configurations use five steps and the same observations. The straight realization isolates coordination on these linear residuals.

The three random seeds are 17011–17013. Their recovery errors are averaged before comparison with the learned model. Each scene contributes one effect averaged over its three nonzero strengths. Descriptive intervals use 20,000 scene-seed bootstrap draws, keeping strengths together. Main-text Table 2 reports learned-versus-random and coordination effects under CG2.

Table 7. Minimax versus mean coordination under systematic bias, with the same learned directions. Positive relative RMSE reduction favors minimax.

Bias strength	CG2	CG0
2/3	+14.36% [10.98, 17.87]	−3.23% [−5.24, −1.26]
1	+12.08% [10.10, 14.26]	−0.72% [−2.62, 1.14]

This comparison supports the joint effect of consensus and refinement under bias. With asymmetric noise, mean coordination has lower recovery error. Local quadratic protection describes the completion subproblem; recovery is measured after the whole trajectory.

Size transfer. Two new biased-graph seeds are evaluated at each dimension with the same check-point and $8 + 8$ direction budget. Learned directions beat equal-rank random directions in all six cases.

Table 8. Mean relative recovery gain of learned completion over equal-rank random completion on the size-transfer instances.

Variables	Versus random
48	+16.401%
192	+33.093%
768	+24.845%

On the same 192-variable instance, graph-isomorphic relabeling changes the mapped-back endpoint by 4.958×10^{-13} . Scaling both view objectives by ten changes it by 4.197×10^{-13} . Scaling only one view changes the objective balance and the endpoint, as expected.

H JACOBIAN BUDGETS

Jacobian budget. Six task types contribute ten small cases each: biased graphs, SE3, multiframe alignment, stereo, known blur and low light. Each reported endpoint comes from one budgeted solve. A joint two-view Jacobian evaluation costs one unit; an individual view costs one half. Internal recomputations count, and cached reuse does not. Recovery is taken from the latest completed iterate within the 21-unit cap, without selecting an iterate by its recovery score.

Table 9. Recovery with a cap of 21 joint Jacobians; VeLO uses 21 joint gradients, reported separately. Ten cases per task.

Task	SETSUNET	HSLM	Ceres LM	Dogbox	VeLO
Biased graph	0.2478	0.1862	0.2587	0.2587	0.7334
SE3	0.0035	0.0039	0.0035	0.0035	0.0061
Multiframe	2.4655	2.0605	2.0399	3.7360	9.5249
Stereo	3.7066	3.9112	3.9361	3.8892	3.9438
Known blur	40.9827	40.9825	40.9825	40.9825	40.3676
Low light	9.9075	8.0177	8.0344	7.9807	8.0846

All five methods have 60 usable records. SETSUNET consumes 21 joint-Jacobian units per case; its graph, SE3 and blur endpoints are at unit 20, while the remaining endpoints are at 21. The inputs use small synthetic problems, stereo crops, three-target multiframe graphs and 16×16 restoration tiles. Solvers returning below the cap retain their returned endpoint and actual count. VeLO receives 21 joint-gradient evaluations with declared horizon 21; its column is not an equal-Jacobian-cost comparison. Incomplete endpoints receive no substitute quality score.

Among these five routes, SETSUNET has the best mean low-light and stereo quality. HSLM has lower biased-graph error and Ceres LM has lower multiframe error. Together with the five-step study, the derivative study evaluates two budgets relevant to local deployment.

Evaluation scope. Cross-method recovery is compared using outer-step and derivative budgets only. The current implementations lack comparable low-level optimization and backend adaptation, so elapsed times are not representative of algorithmic efficiency. The within-SETSUNET whitening comparison in Appendix C separately tests equivalent implementations of the same solver.

The supplementary ZIP includes the minimal inference implementation and compact result tables. Detailed per-case results and derivative counts are available from the authors upon request.